Holistic AI

UCL CENTRE FOR DIGITAL INNOVATION
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Can a web agent learn when a screenshot is worth the cost?

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UCL Centre for Artificial Intelligence — Holistic AI

Choosing the page representation per task could solve up to 16 more tasks in 100 than the best fixed mode — yet none of 8 learned routers beat always using the cheapest mode on both success and cost.

HOW A WEB AGENT SEES A PAGE, AND WHAT WE COMPARED

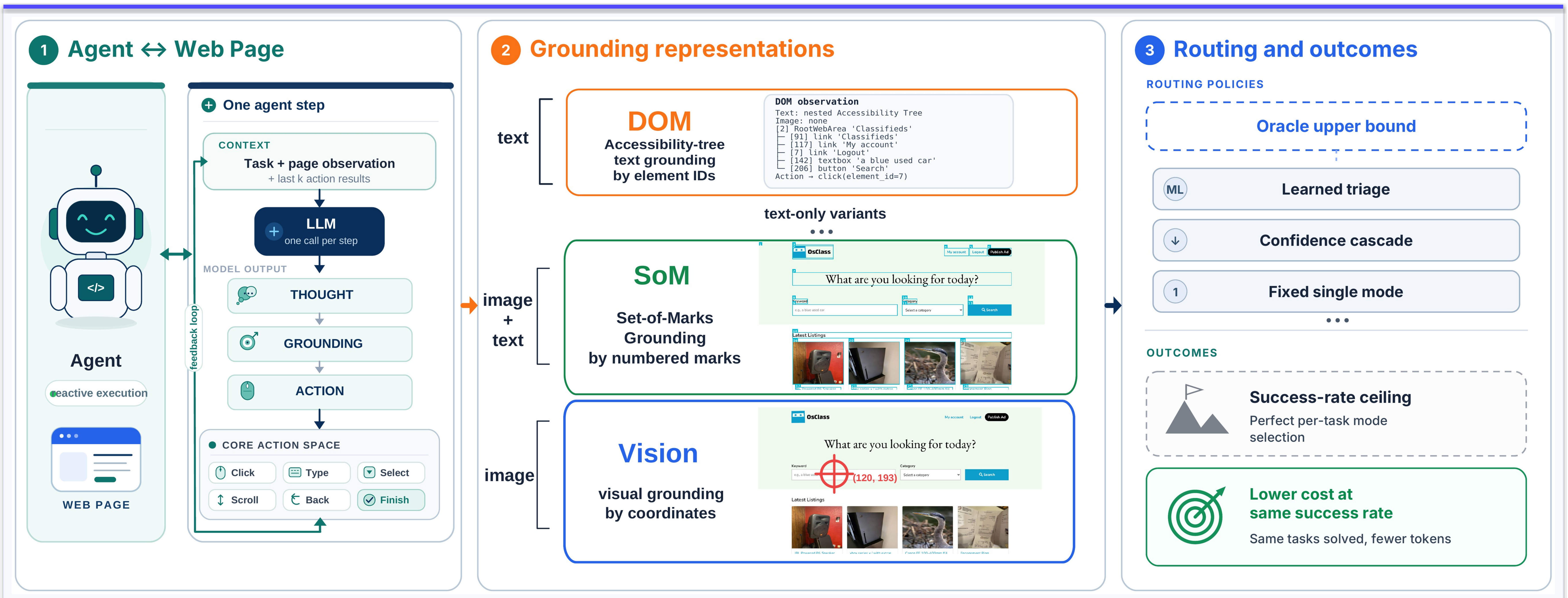

Fig 1. The agent (①) is held fixed — same task, actions, step limit and cost accounting — and only the page encoding it is handed (②) varies: six observation modes, DOM, three text-only variants (P-text, P-prompt, P-SoM), SoM and Vision. Each runs on 8 website–model settings from VisualWebArena and WebArena (8,934 episodes). Three policies (③) are compared: a fixed mode, a hindsight oracle that picks the best mode per task after the fact, and a learned router that must choose before the task runs.

THE PROBLEM

A web agent looks at a page, decides what to click or type, acts, and repeats. Before each step it must be handed some encoding of the page.

That encoding is usually chosen once and paid for at every step: cheap text, an expensive annotated screenshot, or both.

The same page, sent three ways · B0 · classifieds task 0 · first step

DOM	3,314 tokens	text only, no image
SoM	4,335 tokens	text + 143 KB marked image
Vision	3,123 tokens	110 KB screenshot, no text

SoM's text is within 1% of DOM's; nearly all of its extra cost is the image.

Is the screenshot needed at every step — and can the steps that need it be identified cheaply enough to be worth identifying?

WHY IT CANNOT BE LEARNED

A routing label exists only when the agent solves a task. Here the best single mode solves just **2–36%** of tasks, which leaves typically **15–97** usable labels per setting.

The agents that would gain most from routing produce the least supervision to learn it.

Deliberately shrinking the training data confirms scarcity is the mechanism, and prices it: the failing settings would need at least **2.1–4.2×** more tasks than the benchmarks contain — a specification, not an impossibility.

HOW MUCH OF THIS IS NOISE?

Rerunning **one unchanged mode** on the same tasks flips **10–14%** of outcomes and by itself buys **2.0–7.6 pp** of success (B0 · classifieds, six replicated modes, n=224).

Every gain on this sheet is read against that band, not against zero.

RESULTS

+16.35 pp — CEILING, LARGEST OF 8 · VS BEST FIXED MODE

0 of 8 — LEARNED ROUTERS BEAT ALWAYS-CHEAPEST

1 of 8 — HINDSIGHT ORACLES BEAT ALWAYS-CHEAPEST

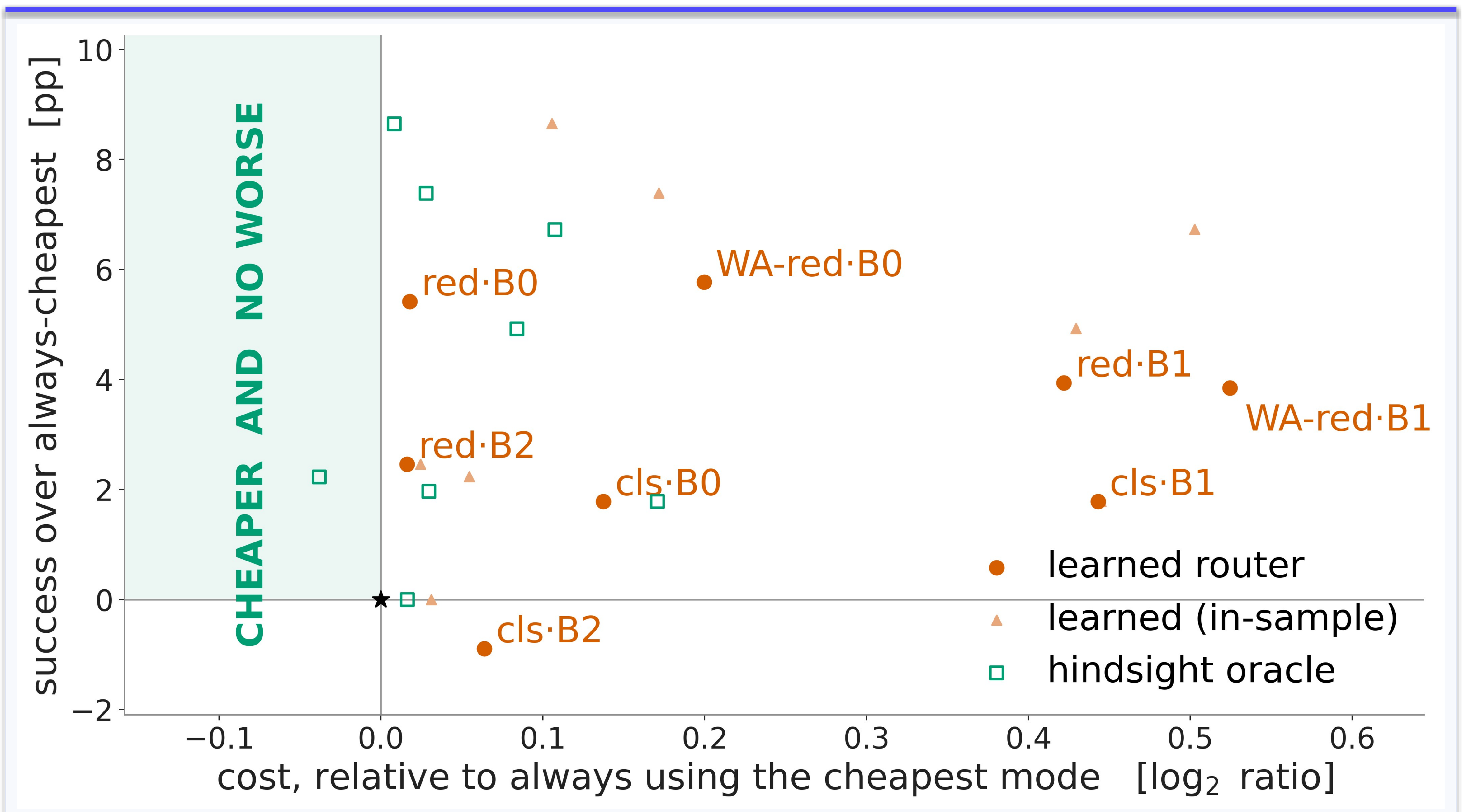

Fig 2. Every policy in every setting against one fixed baseline, **always use the cheapest mode** (★). A win lands in the shaded region: cheaper *and* no worse. Always-cheapest is cheapest on average, not per episode, which is why a few points sit left of it. Nested cross-validation; 10,000 bundle permutations.

The ceiling is real. In hindsight, choosing the mode per task solves **+3.45 to +16.35 pp** more than the best single fixed mode, at 1.6–35.3% lower cost, in 8 of 8 settings.

Nothing we trained wins. 0 of 8 learned routers beat always-cheapest on both success and cost — and even the hindsight oracle does so in only **1 of 8**.

What survives is a bound, not a router. Sending the tasks nobody solves to the cheapest mode saves 9.5–30.6% at identical success in 8 of 8 — against the best-success fixed mode, and plain always-cheapest usually saves more.

TAKEAWAY

Routing is not only a model-selection problem: its learnability depends on the competence of the agent producing the labels. So, in this order: improve the agent, then generate reliable supervision, then learn selective perception.

Measured inside the 2–36% success regime we observed. This conclusion need not hold for stronger agents.

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Paper, code & full results: github.com/Quarkgluonmixture/Cost-Aware-Routing-for-Web-Usage-Agents

Explore all 8 settings →